

Assignment Fundamentals Of Computers

Characteristics Of Computer:

A computer is an electronic machine that receives data, processes it, and gives accurate results. It has several important characteristics which make it useful in all fields:

1. **Speed:** Computers can perform millions of operations in one second. Tasks that may take humans hours or days can be done in just a few seconds.
2. **Accuracy:** Computers give highly accurate results when the input and instructions are correct. Errors usually occur only because of wrong data or wrong instructions.
3. **Automation:** Once a program is started, the computer works automatically without human help until the job is complete.
4. **Storage Capacity:** Computers can store large amounts of data. Information can be saved, retrieved, edited, and reused anytime.
5. **Diligence:** Computers do not get tired or bored. They can work continuously with the same speed and accuracy.
6. **Versatility:** A computer can be used for many tasks—calculations, designing, gaming, communication, data storage, etc.
7. **Reliability:** Computers give consistent results and can be trusted for important work.

Generations Of Computer:

Computer development is divided into generations based on the technology used.

First Generation (1940–1956):

- Used vacuum tubes.
- Very large in size and produced heat.
- Slow and consumed a lot of electricity.
- Example: ENIAC, UNIVAC-I.

Second Generation (1956–1963):

- Used transistors instead of tubes.
- Smaller, faster, and more reliable.
- Consumed less power.
- Example: IBM 7094.

Third Generation (1964–1971):

- Used Integrated Circuits (ICs).
- Faster processing and better storage.
- Smaller and cheaper than earlier generations.

Fourth Generation (1971–Present):

- Use Microprocessors.
- Personal computers were developed.
- Very fast, portable, and affordable.

Fifth Generation (Present and Future):

- Based on Artificial Intelligence, Machine Learning, Robotics.
- Goal is to create computers that think and learn like humans.

Microcomputers:

Microcomputers are small and affordable computers that use a microprocessor as the CPU.

Features of Microcomputers:

- Designed for personal or small business use.
- Low cost and easy to operate.
- Portable and energy efficient.
- Mostly used at homes, schools, offices, and shops.

Examples:

- Desktop computers
- Laptops
- Tablets
- Smartphones

Microcomputers made computers common and available to everyone.

Components Of A Computer System:

A computer system has both physical parts and logical parts. These components work together.

Hardware:

Physical parts of the computer that we can touch. Examples:

- CPU
- Monitor
- Keyboard
- Mouse
- Printer
- Hard disk

Software:

Programs and instructions that tell the computer what to do. Types:

- System Software – Operating systems like Windows, Linux, Android, iOS.
- Application Software – MS Word, browsers, games, calculators.

Different Applications Of Computer:

Computers are used almost everywhere today.

- **Education:** Used for online classes, digital notes, research, and presentations.
- **Business:** Billing, payroll, accounting, inventory management.
- **Healthcare:** Keeping patient records, scanning machines, diagnosis, medical research.
- **Banking:** ATM services, online banking, maintaining accounts.
- **Communication:** Emails, chats, video calls, social media.
- **Entertainment:** Gaming, music, video streaming, animation.
- **Science and Engineering:** Designing machines, simulations, space research.
- **Government:** E-governance, ID services, tax filing, public records.

System Bus And Its Function:

A system bus is a communication pathway that connects various parts of a computer. It transfers data between the CPU, memory, and input/output devices.

Types of System Buses:

1. Data Bus:

- Transfers actual data.
- Bi-directional (two-way movement).

2. Address Bus:

- Carries memory addresses to locate where data should go.
- One-directional.

3. Control Bus:

- Carries control signals like read, write, interrupt, etc.
- Manages and coordinates all activities.

Functions of System Bus:

- Connects CPU with memory and I/O devices.
- Helps in transferring instructions and data.
- Controls the flow of information inside the computer.
- Ensures smooth internal communication.

Input And Output Devices:

Input Devices:

Devices used to enter data and commands into a computer.

Common input devices:

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam
- Touchscreen

Output Devices:

Devices that show the result of processing.

Common output devices:

- Monitor
- Printer
- Speaker
- Headphones
- Projector

Combined Input/Output Devices:

- Touchscreen
- Modem
- VR headsets

CRT Monitors:

CRT stands for Cathode Ray Tube, an older type of display device.

Working Principle:

Inside a CRT, an electron gun shoots beams onto a fluorescent screen. When the beam hits the screen, it lights up and forms images.

Characteristics:

- Bulky and heavy.
- High power consumption.
- Good color and brightness quality.
- Used in old TVs and computers.

Advantages:

- Affordable at the time.
- Strong and durable.
- Good viewing angles.

Disadvantages:

- Large and heavy.
- Heat generation.
- Low energy efficiency.
- Replaced by flat screens like LCD, LED.

Conclusion:

Computers have become a crucial part of modern life. From large vacuum-tube machines to today's microprocessor-based smart devices, computer technology has developed rapidly. Understanding characteristics, generations, components, and applications helps us use computers effectively. Input/output devices and system buses make the system complete, while older devices like CRT monitors show how technology has evolved over time. Overall, computers continue to make work faster, easier, and more accurate in every field.